Bench exercise: take, leave, split

Japanese Joinery by Hand · Episode 4 · Part 1. Cut to the same line three ways and measure what each choice does to the piece. Any fine crosscut saw works: a ryōba's crosscut edge (横挽き), a dōzuki, or a Western backsaw. About 45 minutes.

Safety. Clamp the work; never hold it with your hand in line with the cut. Start the kerf with a few light strokes. Sharp-tool technique is best learned from a teacher, at a bench, in person.

1 · Prepare

1. Take a planed board about 20 × 90 mm, at least 400 mm long, with one straight, square end.
2. From that end, mark three lines across the face at **100 mm**, **200 mm** and **300 mm**, squared from the same reference edge. Use a sharp pencil or a knife, and keep the lines fine and even.
3. Mark the waste side of each line with an ×: the side away from the end you measured from.

2 · Cut three ways

1. **Leave the line** at 100 mm: saw entirely on the waste side, so the whole line stays on the piece.
2. **Split the line** at 200 mm: keep the kerf centered on the line, so a sliver remains on each side.
3. **Take the line** at 300 mm: saw so the kerf just removes the line from the piece.

Cut each piece off and number it 1, 2, 3.

3 · Measure

piece	choice	target	measured	difference
1	leave	100		
2	split	100		
3	take	100		

Measure each offcut's length with a rule or calipers. Leave should be the longest. Split and take differ by only a fraction of a millimetre, and which is shorter depends on whether your kerf is wider than your line. Measure your kerf and your line width and see if that explains it.

4 · Look at the teeth

1. Look along your saw's teeth with a lamp behind. Can you see the set (*asari*), the teeth bent out alternately?
2. If you have a ryōba, compare its two edges: rip teeth like small chisels, crosscut teeth like small knives.

What you should have

- Three pieces, the leave piece longest, and an explanation of the other two from your kerf and line widths.
- A feel for which choice you'd use for a snug fit, and which for a looser one.